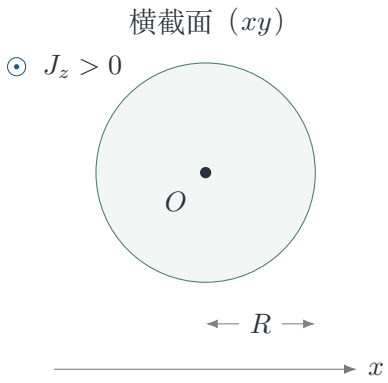
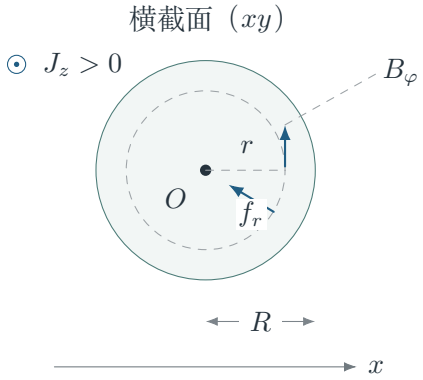
10-W1　均匀圆柱电流的磁挤压



$$J_z = I/(\pi R^2)$$

图 10-13　题图　均匀轴向电流的横截面



$$J_z = I/(\pi R^2)$$

图 10-14　解答分析图　内部环向场产生径向向内的磁力密度

10-W2　环形线圈的截面磁通

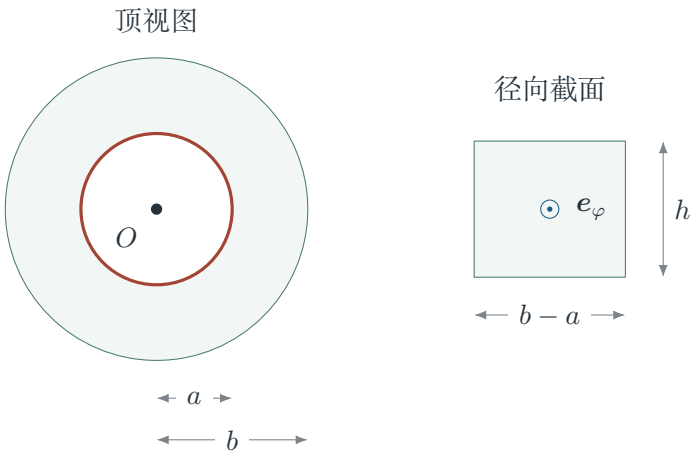


图 10-15　题图　环体顶视图与径向截面，截面法向沿环向

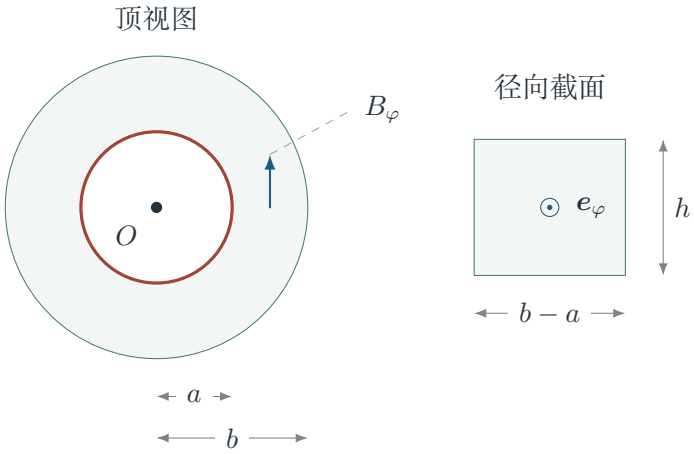


图 10-16　解答分析图　先求环向场，再对截面径向积分

10-W3　偏心孔与主轴处的磁场

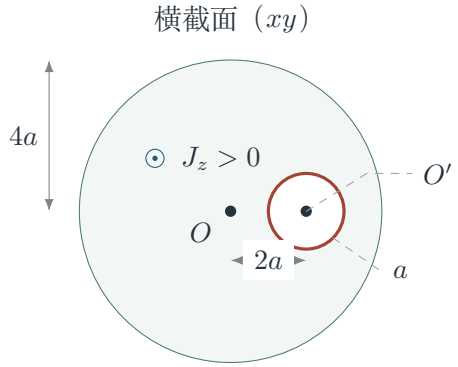


图 10-17　题图　孔半径为 a，孔轴偏移为 $2a$

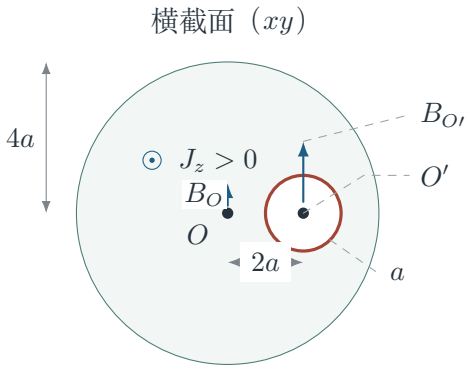


图 10-18　解答分析图　主轴处使用负孔柱的外场；孔内使用两柱内部场

10-W4　半球面上的磁通

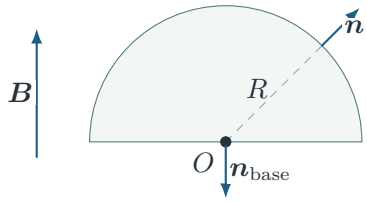
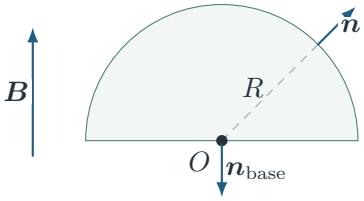


图 10-19　题图　上半球与底圆盘的轴向剖面

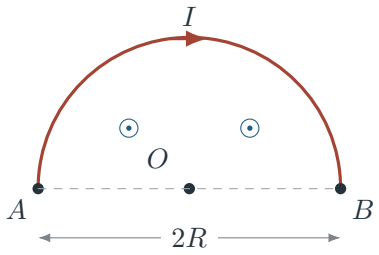


$$\Phi_{\text{hemi}} = +B\pi R^2$$

$$\Phi_{\text{base}} = -B\pi R^2$$

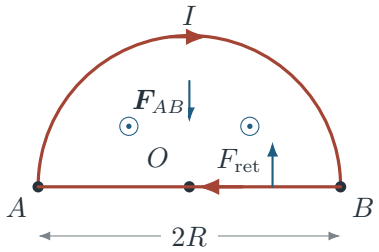
图 10-20　解答分析图　闭面总磁通为零，两部分通量等大反号

10-W5　弯曲导线段在均匀场中的合力



$$\boldsymbol{B} = B_0 \boldsymbol{e}_z$$

图 10-21　题图　电流沿上半圆从左端流向右端，磁场向纸外



$$\boldsymbol{B} = B_0 \boldsymbol{e}_z$$

图 10-22　解答分析图　上半圆合力向下；直径回流受力向上

10-W6　矩形回路受到的非均匀磁力

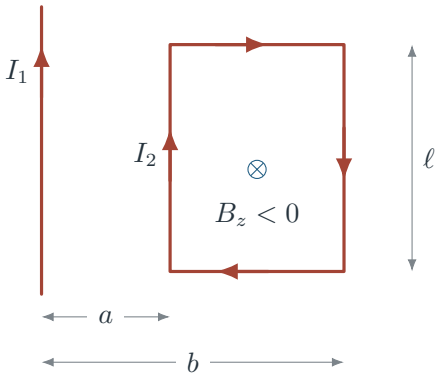


图 10-23　题图　近边电流与源导线同向，远边反向

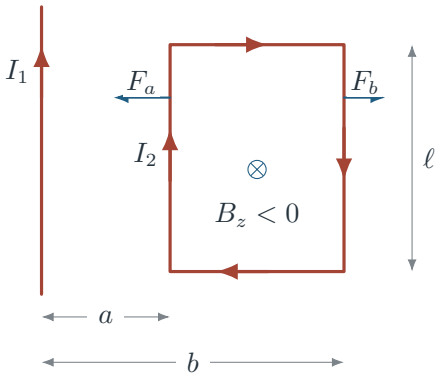


图 10-24　解答分析图　近边吸引大于远边排斥，上下边的力相消

10-W7　刚性线圈的转动与小振动

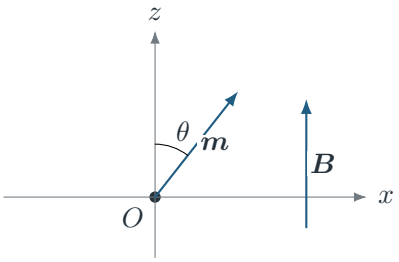
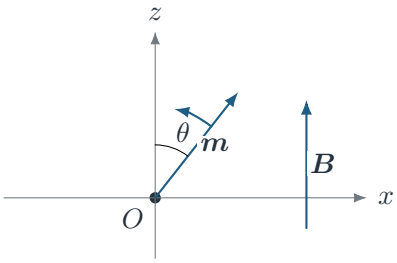


图 10-25　题图　磁矩与场的夹角为 θ



$$\tau_\theta = -mB\sin\theta$$

$$\Omega^2 = mB/\mathcal{I}$$

图 10-26　解答分析图　转动箭头指向同向平衡，小角时为简谐运动

10-W8　线圈转动过程中的机械功

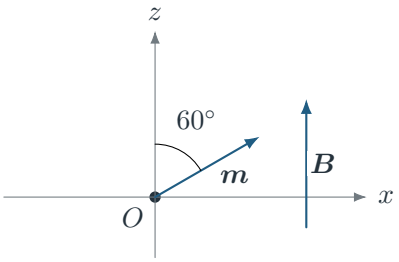
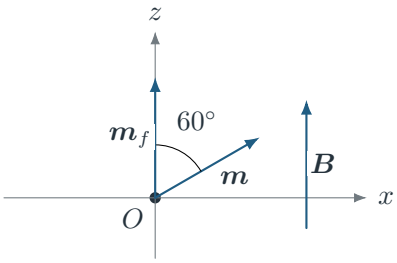


图 10-27　题图　固定磁矩线圈的初始夹角为 60°



$$\theta : \pi/3 \longrightarrow 0$$

$$W_{\mathrm{mag}} = mB/2$$

$$W_{\mathrm{ext}} = -mB/2$$

图 10-28　解答分析图　初态斜向磁矩转至轴向末态，机械功由转角积分求出

10-W9　由螺旋轨迹测量荷质比

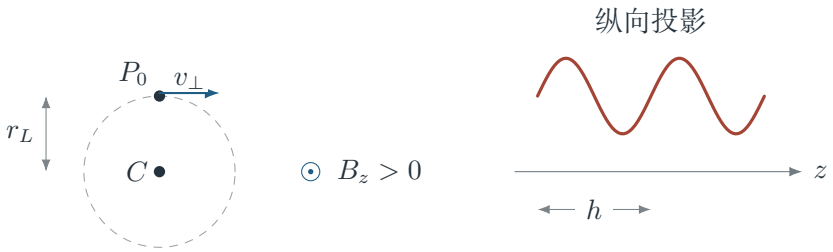


图 10-29　题图　正粒子的横向圆轨道及轴向投影

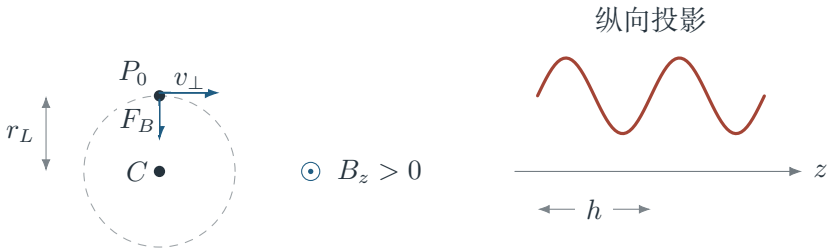


图 10-30　解答分析图　半径确定横向速度，螺距确定纵向速度

10-W10　磁镜中的速度分量与回旋半径

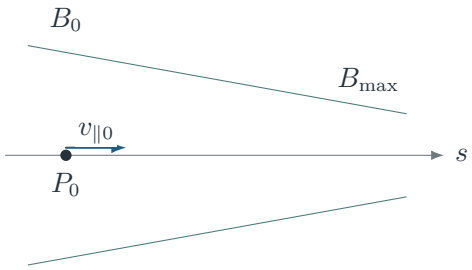


图 10-31　题图　沿轴向逐渐收拢的磁力线，右侧场增强

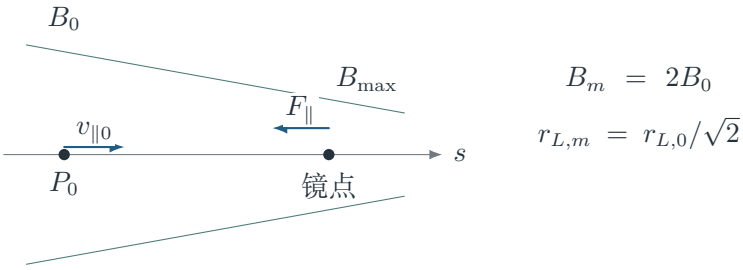


图 10-32　解答分析图　镜力指向弱场，反射时纵向速度为零

10-W11　霍尔测量中的浓度与迁移率

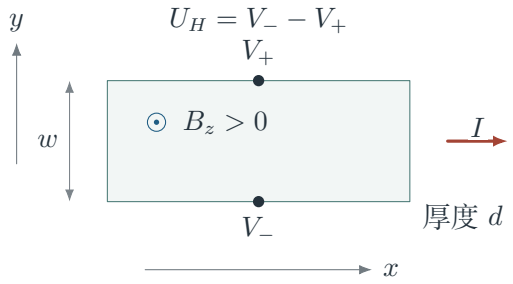


图 10-33　题图　磁场向纸外，电压取下侧减上侧

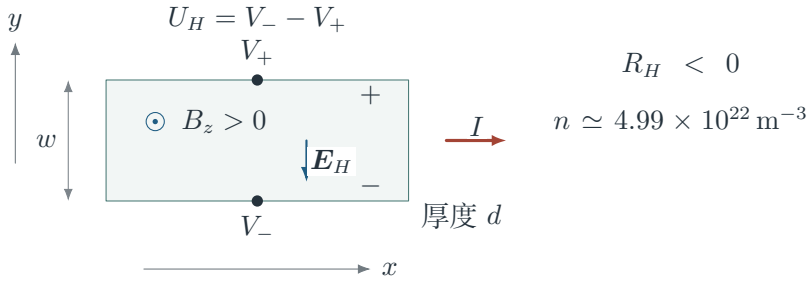


图 10-34　解答分析图　电子在下侧积累，霍尔场向下，电压为负

10-W12　霍尔场与晶格受力

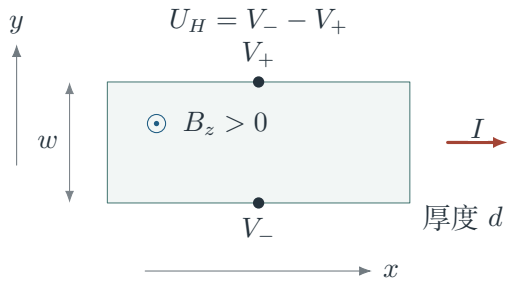


图 10-35　题图　电子金属条的电流、场和接线方向

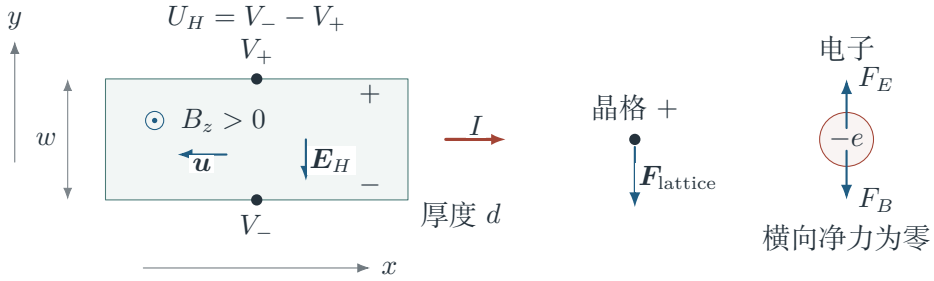
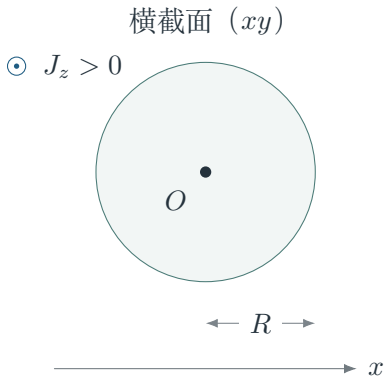


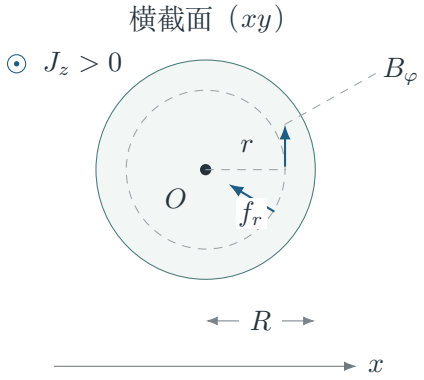
图 10-36　解答分析图　电子横向净力为零，晶格经霍尔场受向下的力

10-T1　非均匀电流的压力分布



$$J_z = J_0(1 - r^2/R^2)$$

图 10-37　题图　中心电流密度较大、表面为零的圆柱



$$J_z = J_0(1 - r^2/R^2)$$

图 10-55　解答分析图　累计电流给出环向场，压力梯度平衡径向磁力

10-T2　带背景场的无限长螺线管

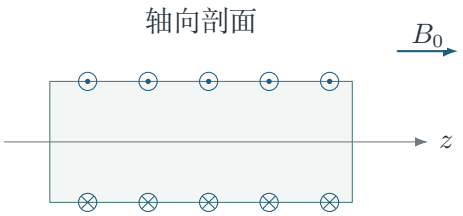
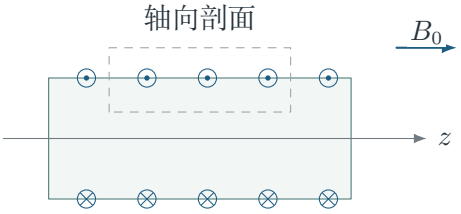


图 10-38　题图　环向面电流及远处给定的轴向背景场

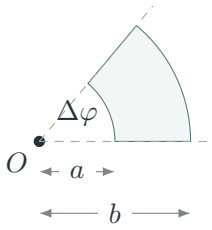


$$B_{\text{in}} - B_{\text{out}} = \mu_0 K$$

$$B_{\text{out}} = B_0$$

图 10-56　解答分析图　穿壁矩形环路只确定两侧场之差

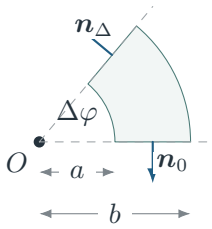
10-T3　环体小扇区的闭合磁通



顶视图；高度 h

$B_\varphi = C/r$

图 10-39　题图　环体中一个有限角度小扇区的顶视图

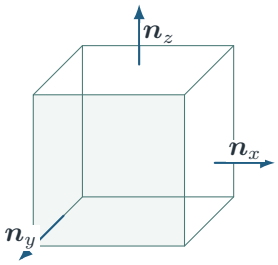


顶视图；高度 h

$B_\varphi = C/r$

图 10-57　解答分析图　两张径向截面的通量相消，另四面通量为零

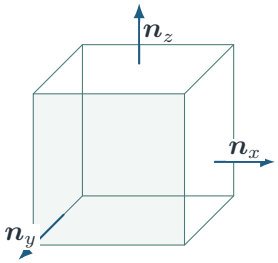
10-T4　线性磁场的闭面条件



$$B_x = \alpha x$$
$$B_y = \beta y$$
$$B_z = \gamma z$$

边长 $2a$

图 10-40　题图　立方体的三个正向面及相应外法向



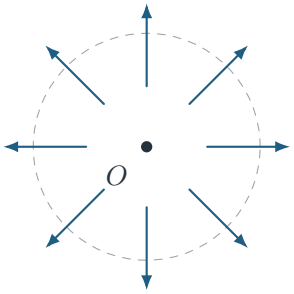
$$B_x = \alpha x$$
$$B_y = \beta y$$
$$B_z = \gamma z$$

边长 $2a$

$$\alpha + \beta + \gamma = 0$$

图 10-58　解答分析图　每一对相反面通量同号，六面总和由散度决定

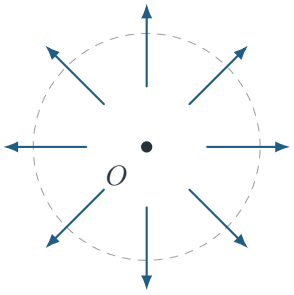
10-T5　径向候选场与源点奇性



候选场 $\boldsymbol{B}=C\boldsymbol{e}_r/r^2$

球面半径 a

图 10-41　题图　待检验的径向候选场，图示取 $C>0$



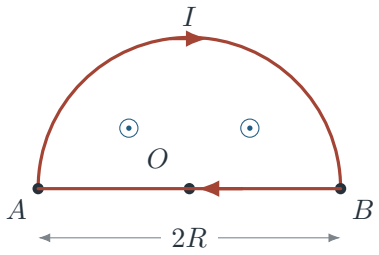
候选场 $\boldsymbol{B}=C\boldsymbol{e}_r/r^2$

球面半径 a

$\Phi=4\pi C$

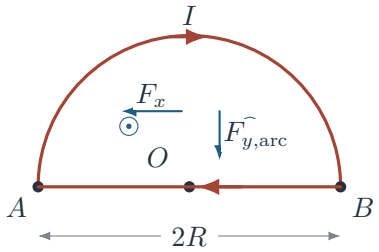
图 10-59　解答分析图　穿出球面的通量不为零，原点奇性不能遗漏

10-T6　线性梯度场中的半圆回路



$$B_z = B_0 + gx$$

图 10-42　题图　顺时针半圆闭合回路，磁场向纸外且沿 x 变化



$\otimes\ m_z < 0$

$$B_z = B_0 + gx$$

图 10-60　解答分析图　分析示意取 $g > 0$；均匀部分的力相消，梯度部分合力向左

10-T7　三根平行导线的横向稳定性

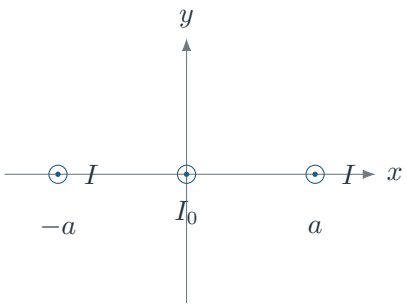


图 10-43　题图　三根平行导线的横截面，第三根方向以 $I_0 > 0$ 示意

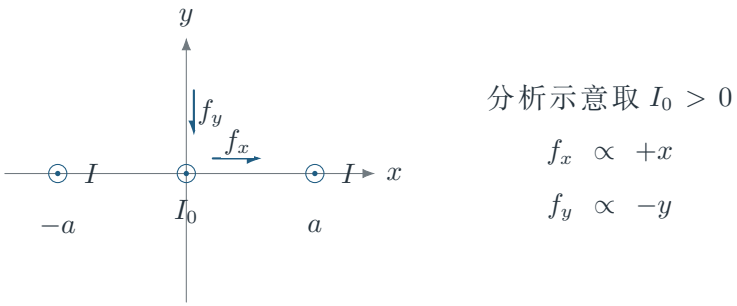


图 10-61　解答分析图　中心线向右位移失稳，向上位移有向下回复力

10-T8　扭转弹簧约束的反向磁矩

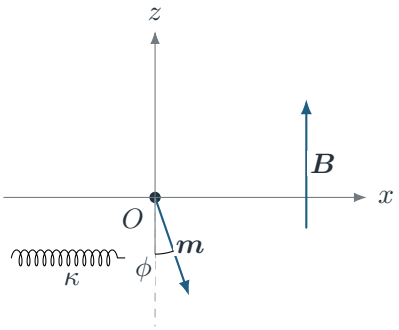
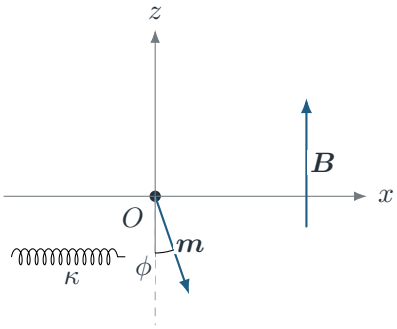


图 10-44　题图　扭转弹簧的原平衡磁矩反向于外场



$U = \kappa\phi^2/2 + mB\cos\phi$

$\kappa = mB$ 时四阶项为正

图 10-62　解答分析图　比较弹簧的回复力矩与反向磁矩的失稳力矩

10-T9　满足无散条件的磁场梯度

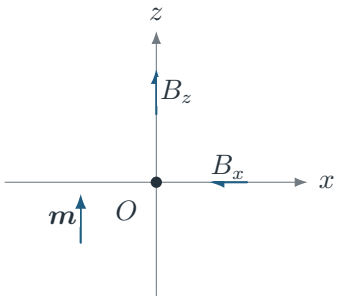
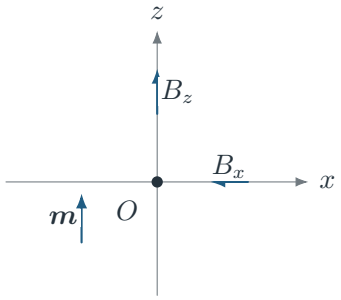


图 10-45　题图　线性场的 xz 截面；轴向增强伴随横向收拢



图示 $g>0$, $\boldsymbol{m}=m\boldsymbol{e}_z$

$$F_z = mg$$

$$\boldsymbol{\tau} = 0$$

图 10-63　解答分析图　图示同向磁矩受到向上的梯度力

10-T10　有限宽度磁场中的出射位置

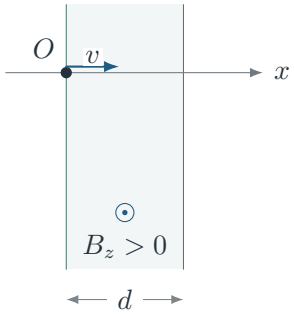


图 10-46　题图　正粒子从左边界射入有限宽度磁场

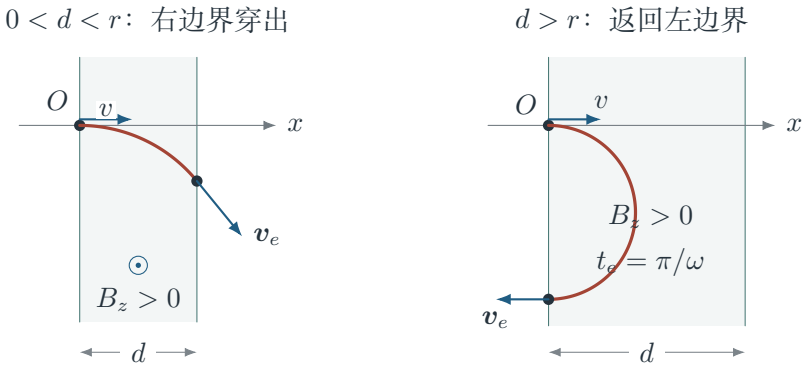


图 10-64　解答分析图　并列右侧穿出与返回左边界两种轨迹；相切临界另判

10-T11　回旋周期的相对论修正

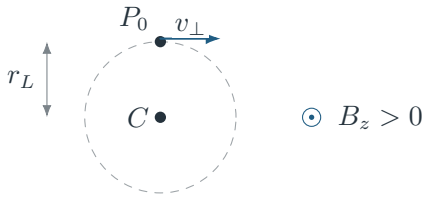


图 10-47　题图　圆轨道与周期关系，方向示意取正电荷

$$T/T_{\mathrm{nr}} \;=\; \gamma$$

$$p \;=\; |q|Br_L$$

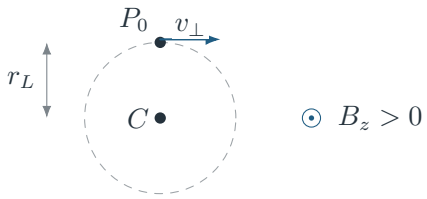


图 10-65　解答分析图　高速时由动量求半径，周期含相对论因子

10-T12　平行电场中的变螺距轨迹

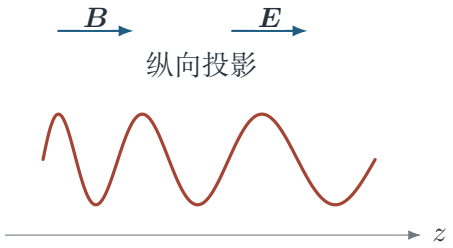


图 10-48　题图　图示 $qE>0$，相邻回旋周期的轴向位移增加

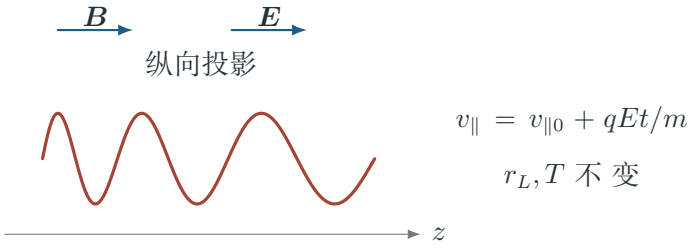


图 10-66　解答分析图　横向回旋不变，纵向匀加速导致螺距变化

10-T13　两端不对称的磁镜

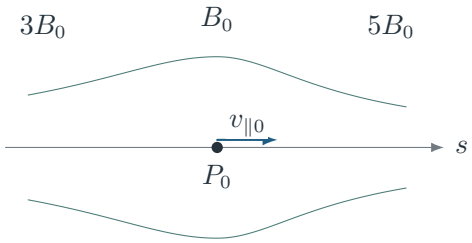
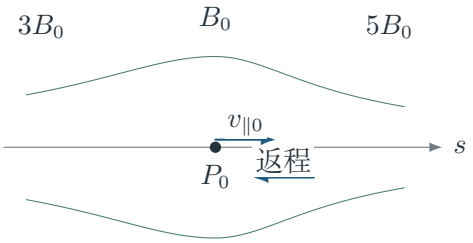


图 10-49　题图　左右端的最大场不同，中部为弱场



$B_m = B_0/\sin^2\alpha$

$35°$：右侧反射、左侧逸出

图 10-67　解答分析图　较弱的左端限制双端约束的入射角

10-T14　近轴磁阱中的往返周期

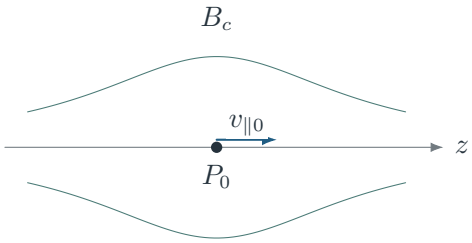
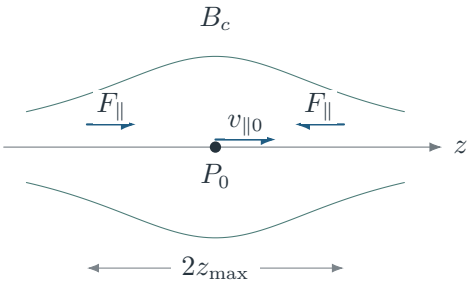


图 10-50　题图　近轴磁阱的磁力线示意，中部场最弱



$$\ddot{z} + \Omega_b^2 z = 0$$

$$\Omega_b = v_{\perp 0}/L$$

图 10-68　解答分析图　导引中心在两个镜点间往返，回复力与 z 成正比

10-T15　反转电流与磁场的霍尔读数

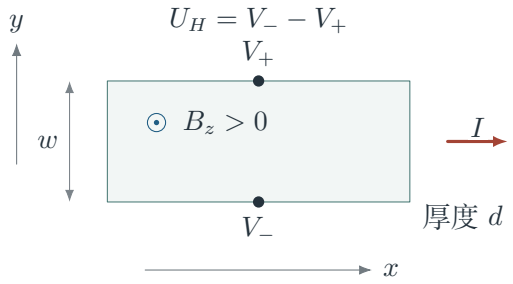


图 10-51　题图　横向电压接线及电流、磁场的正方向

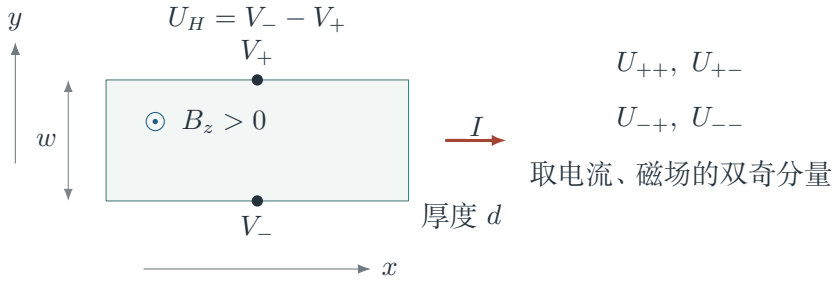


图 10-69　解答分析图　两种独立反转可分离霍尔项和触点偏差

10-T16　双载流子的霍尔抵消

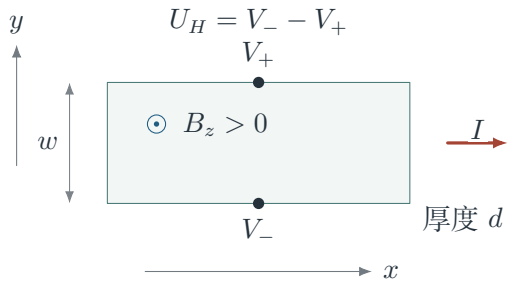


图 10-52　题图　两类载流子共存，电流仍沿正 x

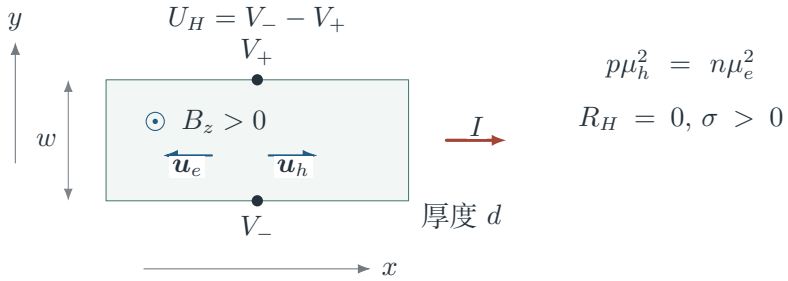


图 10-70　解答分析图　横向总电流为零，两类霍尔项可相互抵消

10-T17　直导线旁的霍尔条与支承力（限时综合）

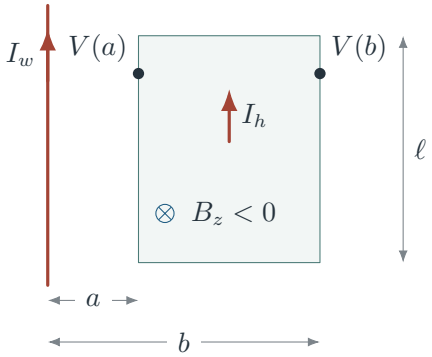


图 10-53　题图　霍尔条电流平行于源导线，磁场随横向距离变化

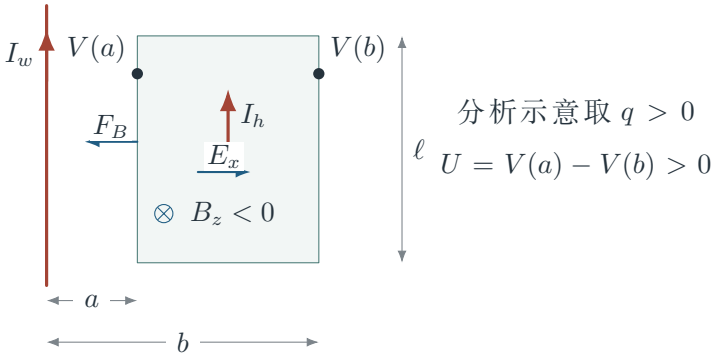


图 10-71　解答分析图　霍尔电压与总磁力均须对宽度积分

10-T18　霍尔测场与粒子磁镜（限时综合）

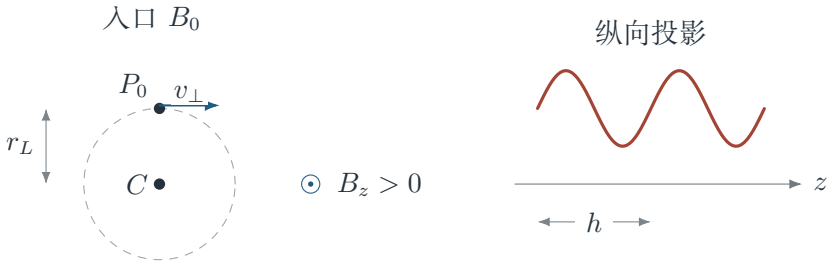


图 10-54　题图　入口回旋半径和螺距与霍尔探头共同确定参数

$$B_0 = U_0 d_s/(|R_H| I_s)$$

$$B_m/B_0 = 1 + h^2/(4\pi^2 r^2)$$

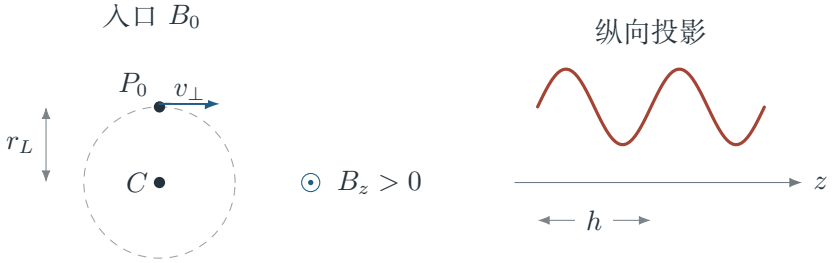


图 10-72　解答分析图　先标定入口场，再用能量与绝热不变量求镜点